Natalyx MVP Fixes

Work Package 07: Commercial configuration, enrollment, and promotion

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1 Work Package 07: Commercial configuration, enrollment, and promotion

Review draft only. Ownership and sequencing remain recommendations until Luke and Alan approve the plan together.

Plan	Value
Work package	WP07
Plan file	`WP07 - Commercial configuration, enrollment, and promotion.md`
Proposed PRs	P07-01, P07-02, P07-03, P07-04
Inspection snapshot	2026-07-12 19:30 America/New_York

1.1 Purpose and outcome

Problem. The business model requires versioned price/terms, explicit commitment, and one precise processor-neutral event before workspace-to-case promotion; none is currently authoritative.

Why this package exists. It creates a reviewable boundary so the programme can progress without mixing product, data, security, infrastructure and rollout concerns in one oversized change.

Completed outcome. Versioned commercial configuration and acceptances, a separate commitment/enrollment lifecycle, one processor-neutral byos.workspace_promotion_authorized.v1 event, and exactly-once paid-case creation by its consumer.

Explicit non-goals.

- No processor-specific webhook
- No automated unresolved pricing policy
- No activation before onboarding evidence
- No browser-redirect authority

1.2 Verified current-state evidence

Classification is explicit: **verified** is observed code/PR/live behavior; **verified limitation** records an evidence gap; **design authority** is an approved planning source; **design decision** is proposed future behavior. None is silently treated as another.

ID	Class	Repository / ref	File, construct or location	Finding
E-025	design authority	business model / Natalyx_Payment_and_Business_Model_BYOS_MVP.docx v1.0	Sections 3-12 and implementation tables	Node 0 Free must complete the match; Complete is optional/creditable; $8,900 JMF and $750 enrollment anchor the configurable model; promotion follows explicit commitment and funding, not willingness or redirect.

1.3 Architecture and contracts

- Every commitment, invoice, payment, credit, refund, and promotion records the activated commercial-configuration version.
- Only the app-owned commitment/commercial domain emits byos.workspace_promotion_authorized.v1 after durable funding/terms/credit validation; a processor callback or redirect never emits it.
- The consumer creates/fetches one enrollment_pending case and restricted IP membership; GC case access follows the paid-case invitation contract.

Contract area	Package rule
Ownership boundary	`app`; repository-specific PRs remain separate.
Entities/state	`case`, `case_invitation`, `journey_commitment`, `node0_workspace`
Authorization	Authenticated subject/service plus entity membership, role, purpose, consent and expected revision; query parameters, links and UI state are never authority.
API/service	Read active config and record acceptance commands. Create, accept, cancel, inspect enrollment. Internal event producer only. Internal event handler only.
Storage	Commercial configuration and acceptance ledgers. Commitment events/current projection. Promotion ledger and source-workspace unique constraint.

Contract area	Package rule
Audit	Use the common transition envelope and append-only source/event identifiers; redact tokens, raw documents, prompts and sensitive payloads.
Failure	Fail closed for authorization, provenance, money, document and PHI boundaries; expose a bounded actionable error and preserve retry/repair evidence.
Observability	Correlation, policy/schema/config version, transition and dependency result metrics without PHI/PII payload logging.
Compatibility	Additive/dual-compatible changes, explicit version pins, old/new revision window and named forward repair.
Privacy/PHI	Synthetic only through A2; approved PII only at B1; PHI remains blocked until formal B2.

1.3.1 Applicable state machines

1.3.2 Node 0 workspace (`node0_workspace`)

Owner: **natalyx-app**. States: `draft`, `active`, `match_ready`, `promotion_pending`, `promoted`, `archived`.

Transition: `none` -> `draft`

- **Actor:** authenticated IP
- **Evidence:** first explicit Node 0 start
- **Idempotency key:** `workspace:{owner}:create`
- **Invalid transitions:** Existing active workspace returns existing
- **Audit event:** `workspace_created`
- **Retry:** Unique owner/idempotency key
- **Recovery:** Forward repair orphan membership

Transition: `draft` -> `active`

- **Actor:** IP
- **Evidence:** minimum workspace consent/profile
- **Idempotency key:** `workspace:{id}:activate:{revision}`
- **Invalid transitions:** Missing consent/owner
- **Audit event:** `workspace_activated`
- **Retry:** Return current active revision
- **Recovery:** Correct inputs and retry

Transition: `active` -> `match_ready`

- **Actor:** IP + selected GC evidence policy
- **Evidence:** selected candidate + required provisional/GC-authored fields
- **Idempotency key:** `workspace:{id}:ready:{revision}`

- **Invalid transitions:** Critical unknown/decline/revoked claim
- **Audit event:** `workspace_match_ready`
- **Retry:** No duplicate event
- **Recovery:** Reopen active with reason revision

Transition: `match_ready -> promotion_pending`

- **Actor:** promotion event consumer
- **Evidence:** valid byos.workspace_promotion_authorized.v1
- **Idempotency key:** `promotion:{commitment}:v1`
- **Invalid transitions:** No authorization; already promoted; stale config
- **Audit event:** `workspace_promotion_requested`
- **Retry:** Replay resumes/finds same promotion
- **Recovery:** Forward-repair incomplete transaction

Transition: `promotion_pending -> promoted`

- **Actor:** promotion service
- **Evidence:** paid case + manifest committed
- **Idempotency key:** `promotion:{commitment}:v1`
- **Invalid transitions:** Projection invariant/privacy failure
- **Audit event:** `workspace_promoted`
- **Retry:** Return same case ID
- **Recovery:** Repair from immutable manifest; never duplicate

Transition: `draft|active|match_ready -> archived`

- **Actor:** IP/operator per policy
- **Evidence:** explicit archive + retention policy
- **Idempotency key:** `workspace:{id}:archive:{request}`
- **Invalid transitions:** Promotion pending/promoted
- **Audit event:** `workspace_archived`
- **Retry:** Idempotent
- **Recovery:** Restore only through new audited revision if policy permits

1.3.3 Commercial commitment (`journey_commitment`)

Owner: **natalyx-app commercial domain**. States: `draft`, `accepted`, `funding_pending`, `funded`, `cancelled`, `expired`.

Transition: `none -> draft`

- **Actor:** IP
- **Evidence:** active workspace + displayed config/terms
- **Idempotency key:** `commitment:{workspace}:{config}`
- **Invalid transitions:** No active config/workspace
- **Audit event:** `commitment_drafted`

- **Retry:** Return existing draft
- **Recovery:** Create new draft after expiry/version change

Transition: `draft -> accepted`

- **Actor:** IP
- **Evidence:** terms/privacy consent + payment path
- **Idempotency key:** `commitment:{id}:accept:{terms_hash}`
- **Invalid transitions:** Stale/missing acceptance
- **Audit event:** `commitment_accepted`
- **Retry:** Return accepted
- **Recovery:** New acceptance revision if version changes

Transition: `accepted -> funding_pending`

- **Actor:** commercial service
- **Evidence:** enrollment invoice/payment intent created
- **Idempotency key:** `commitment:{id}:funding`
- **Invalid transitions:** No invoice or invalid amount
- **Audit event:** `funding_requested`
- **Retry:** Resume same invoice
- **Recovery:** Cancel/reissue under audited adjustment

Transition: `funding_pending -> funded`

- **Actor:** payment domain
- **Evidence:** net settled enrollment funding meets versioned requirement
- **Idempotency key:** `enrollment-funding:{id}:v1`
- **Invalid transitions:** Redirect/pending/failed/reversed evidence
- **Audit event:** `commercial.enrollment_funding_confirmed.v1`
- **Retry:** Exactly-once event/outbox replay
- **Recovery:** Reconcile inbox/ledger; no browser repair

Transition: `draft|accepted|funding_pending -> cancelled`

- **Actor:** IP/authorized operator
- **Evidence:** explicit cancellation and refund policy
- **Idempotency key:** `commitment:{id}:cancel:{request}`
- **Invalid transitions:** Already funded/promoted uses case cancellation rail
- **Audit event:** `commitment_cancelled`
- **Retry:** Idempotent
- **Recovery:** New commitment required

Transition: `draft|accepted -> expired`

- **Actor:** system

- **Evidence:** offer/terms TTL or config deactivation
- **Idempotency key:** `commitment:{id}:expire`
- **Invalid transitions:** No expiry after funding
- **Audit event:** `commitment_expired`
- **Retry:** Idempotent
- **Recovery:** Draft new version

1.3.4 Paid case lifecycle (`case`)

Owner: **natalyx-app**. States: `enrollment_pending`, `active`, `paused`, `closed`, `cancelled`.

Transition: `none` -> `enrollment_pending`

- **Actor:** promotion service
- **Evidence:** byos.workspace_promotion_authorized.v1 + projection manifest
- **Idempotency key:** `promotion:{commitment}:v1`
- **Invalid transitions:** No authorization/duplicate workspace
- **Audit event:** `paid_case_created`
- **Retry:** Return same case
- **Recovery:** Forward-repair same promotion

Transition: `enrollment_pending` -> `active`

- **Actor:** authorized coordinator/domain service
- **Evidence:** approved onboarding/compatibility/intake/education evidence
- **Idempotency key:** `case:{id}:activate:{evidence_hash}`
- **Invalid transitions:** Missing/revoked evidence or GC consent
- **Audit event:** `case_activated`
- **Retry:** No duplicate activation/invoice
- **Recovery:** Return pending with unmet checklist

Transition: `active` -> `paused`

- **Actor:** authorized commercial/operations policy
- **Evidence:** audited non-critical pause reason
- **Idempotency key:** `case:{id}:pause:{event}`
- **Invalid transitions:** Safety-critical access cannot be paused
- **Audit event:** `case_paused`
- **Retry:** Idempotent
- **Recovery:** Resume after evidence; preserve entitled docs

Transition: `paused` -> `active`

- **Actor:** authorized operator/domain service
- **Evidence:** pause reason resolved
- **Idempotency key:** `case:{id}:resume:{event}`
- **Invalid transitions:** Unresolved safety/authorization block

- **Audit event:** `case_resumed`
- **Retry:** Idempotent
- **Recovery:** Escalate manual review

Transition: `active|paused -> closed`

- **Actor:** authorized coordinator
- **Evidence:** birth/close or completed termination handoff evidence
- **Idempotency key:** `case:{id}:close:{evidence_hash}`
- **Invalid transitions:** Open critical obligations
- **Audit event:** `case_closed`
- **Retry:** No-op if same evidence
- **Recovery:** Reopen only by exceptional audited event

Transition: `enrollment_pending|active|paused -> cancelled`

- **Actor:** authorized operator under policy
- **Evidence:** withdrawal/non-clearance/cancellation + financial disposition
- **Idempotency key:** `case:{id}:cancel:{request}`
- **Invalid transitions:** Cannot erase records/refund ledger
- **Audit event:** `case_cancelled`
- **Retry:** Idempotent
- **Recovery:** Replacement is a new case/workspace policy event

1.3.5 Paid-case membership invitation (`case_invitation`)

Owner: **natalyx-app**. States: `issued`, `pending_identity`, `identity_bound`, `mismatch_pending`, `issuer_confirmed`, `consented`, `redeemed`, `declined`, `revoked`, `expired`, `superseded`.

Transition: `none -> issued`

- **Actor:** authorized IP case member
- **Evidence:** enrollment_pending/active case + recipient binding
- **Idempotency key:** `case-invite:{case}:{role}:{generation}`
- **Invalid transitions:** Self/member/active-GC/unauthorized
- **Audit event:** `case_invite_issued`
- **Retry:** Return current; supersede prior live GC invite
- **Recovery:** Issue corrected invite

Transition: `issued -> pending_identity`

- **Actor:** recipient
- **Evidence:** valid addressed token landing
- **Idempotency key:** `case-invite:{id}:land`
- **Invalid transitions:** Expired/revoked
- **Audit event:** `case_invite_opened`
- **Retry:** Safe repeat

- **Recovery:** Authenticate with return_to

Transition: `pending_identity -> identity_bound|mismatch_pending`

- **Actor:** auth provider
- **Evidence:** verified subject/email
- **Idempotency key:** `case-invite:{id}:bind:{subject}`
- **Invalid transitions:** Terminal/duplicate/different subject
- **Audit event:** `case_invite_bound|mismatch`
- **Retry:** CAS protects terminal
- **Recovery:** Issuer confirmation or corrected invite

Transition: `mismatch_pending -> issuer_confirmed`

- **Actor:** authorized issuer
- **Evidence:** explicit reviewed mismatch confirmation
- **Idempotency key:** `case-invite:{id}:confirm:{issuer}`
- **Invalid transitions:** Issuer unauthorized/recipient changed
- **Audit event:** `case_invite_mismatch_confirmed`
- **Retry:** CAS no-op
- **Recovery:** Revoke and reissue

Transition: `identity_bound|issuer_confirmed -> consented`

- **Actor:** invitee
- **Evidence:** paid-case role/privacy consent version
- **Idempotency key:** `case-invite:{id}:consent:{version}`
- **Invalid transitions:** Stale/missing consent
- **Audit event:** `case_invite_consented`
- **Retry:** Idempotent
- **Recovery:** Present current consent

Transition: `consented -> redeemed`

- **Actor:** invitation service
- **Evidence:** single transaction creates person/party/membership
- **Idempotency key:** `case-invite:{id}:redeem`
- **Invalid transitions:** Existing member/active GC/expired/revoked
- **Audit event:** `case_invite_redeemed`
- **Retry:** Return same membership
- **Recovery:** Rollback entire transaction; forward repair from invite

Transition: `issued|pending_identity|identity_bound|mismatch_pending|issuer_confirmed|consented -> declined|revoked|expired|superseded`

- **Actor:** invitee/issuer/system

- **Evidence:** explicit decline/revoke/TTL/new generation
- **Idempotency key:** `case-invite:{id}:terminal:{event}`
- **Invalid transitions:** No terminal overwrite
- **Audit event:** `case_invite_terminal`
- **Retry:** Idempotent CAS
- **Recovery:** Fresh invitation only

1.3.6 Data authorship/provenance rules

- Every assertion/revision records subject_ref, semantic_key, author_subject/person/role, authorship_class, provenance, verification_status, visibility, purpose, source_ref, derived_from, supersedes_id, promotion_event_id, policy version and immutable timestamps.
- Class 1 defaults IP-private; class 2 defaults IP-private/internal and visibly says reported by the intended parent, not confirmed by the GC.
- Class 3 is GC-authored and shared only per field/purpose; class 4 is GC-only private; class 5 requires an evidence-linked role-gated reviewer/provider rail.
- GC confirmation appends a class-3 revision referencing the class-2 assertion. Relationship answers can coexist; disagreement is not an overwrite.
- Unselected candidates, raw health assertions, ranking scores and private IP notes do not promote. Derived values inherit the weakest source status.
- Copying, promotion, generic proposal approval, IP assertion or AI inference can never create Natalyx-reviewed, provider-confirmed, medically cleared, legally cleared or insurance-verified truth.

1.3.7 Sole promotion-authorizing event

`event_type: byos.workspace_promotion_authorized.v1`

Producer: natalyx-app commitment/commercial domain after deterministic validation. Consumer: natalyx-app promotion service.

Payload:

```
event_id
idempotency_key
workspace_id
commitment_id
owner_person_id
owner_subject
commercial_config_version
terms_acceptance_ids[]
payment_plan_id
funding_evidence.kind
funding_evidence.commercial_event_id
funding_evidence.settled_amount_minor
funding_evidence.currency
credit_allocation_ids[]
initial_case_state=enrollment_pending
occurred_at
```

`correlation_id`
`causation_id`
`schema_version`

Authorization conditions:

- Workspace is active/match-ready, unarchived and not already promoted.
- Commitment references the same workspace, has current terms/privacy acceptances and is in funding_pending/funded per the approved state contract.
- Payment path and activated commercial configuration are pinned.
- Required enrollment funding has a verified settled ledger entry or separately approved financing/manual-funding evidence; a processor webhook is first normalized and cannot itself authorize promotion.
- Credit allocation is non-negative, reproducible and cannot exceed the applicable obligation.
- No cancellation, compliance hold, reversal that invalidates funding, or existing promotion is present.

Unique constraints:

- one promotion per workspace
- one case per promotion event
- one result per idempotency key
- one primary owner membership per created case

No browser success/return route and no raw processor callback can emit or substitute for this event.

1.4 Explicit PR sequence

Order	ID	Title	Repository	Dependencies	Gate	Local preparation	Merge readiness
1	P07-01	Add versioned commercial configuration and terms acceptance	app	P02-03	A2	Not ready; named PR, decision or programme-acceptance prerequisites remain	Blocked pending PR dependencies P02-03; Gate A0 active and verified; Alan review

Order	ID	Title	Repository	Dependencies	Gate	Local preparation	Merge readiness
2	P07-02	Implement commitment and enrollment-pending lifecycle	app	P07-01, P08-01	A2	Not ready; named PR, decision or programme-acceptance prerequisites remain	Blocked pending PR dependencies P07-01, P08-01; Gate A0 active and verified; Alan review
3	P07-03	Emit processor-neutral workspace promotion authorization	app	P04-02, P07-02, P08-02	A2	Not ready; named PR, decision or programme-acceptance prerequisites remain	Blocked pending PR dependencies P04-02, P07-02, P08-02; Gate A0 active and verified; Alan review
4	P07-04	Promote workspace exactly once from authorization event	app	P03-03, P04-02, P07-03	A2	Not ready; named PR, decision or programme-acceptance prerequisites remain	Blocked pending PR dependencies P03-03, P04-02, P07-03; Gate A0 active and verified; Alan review

1.4.1 P07-01 - Add versioned commercial configuration and terms acceptance

Field	Plan
Repository / base	`app` (`main`)
Gate / size	A2 / Large
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P02-03; Gate A0 active and verified; Alan review
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Dependencies	P02-03
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	Payment/business-model DOCX

1.4.1.1 Objective

Store activated price books, milestone schedules, Node 0 credit rules, and immutable terms/privacy acceptance versions without scattering literals.

1.4.1.2 Scope

- Immutable activated config versions
- JMF/deposit/discount/milestones/credit policy
- Terms and privacy acceptance records

1.4.1.3 Explicit non-goals

- No unresolved Node 0 price value
- No payment provider

1.4.1.4 Expected files or modules

- `migrations/versions/*_commercial_configuration.py`
- `app/domain/commercial_config.py`
- `app/storage/commercial_config.py`
- `tests/python/test_commercial_config.py`

1.4.1.5 Change map

Area	Planned change
Schema/model	Commercial configuration and acceptance ledgers.
API/routes	Read active config and record acceptance commands.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	Apply through the named versioned migration; no ad-hoc startup DDL. Backfill is none unless explicitly listed in scope.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

1.4.1.6 Tests and validation

Objective tests:

- Activation immutability
- currency/amount validation
- acceptance version
- no retroactive mutation

Local validation:

- `python -m unittest discover -s tests/python`

- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate **A2** on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

1.4.1.7 Diagnostics and observability Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

1.4.1.8 Rollout Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

1.4.1.9 Rollback or forward repair Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

1.4.1.10 Dependencies and parallel work Hard dependencies: P02-03. Parallel opportunities: P07-03, P07-04

1.4.1.11 Objective acceptance criteria

- ☐ Store activated price books, milestone schedules, Node 0 credit rules, and immutable terms/privacy acceptance versions without scattering literals.
- ☐ Every listed objective test passes locally and at the required gate where applicable.
- ☐ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- ☐ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- ☐ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

1.4.1.12 Ownership and review

Role	Recommendation
Work-package DRI	Luke
Primary implementer	Luke (proposed; pending Alan review)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
PR-level ownership exception	None
Required expertise	commercial systems, PostgreSQL
Higher-capability review	Mandatory before merge

1.4.2 P07-02 - Implement commitment and enrollment-pending lifecycle

Field	Plan
Repository / base	`app` (`main`)
Gate / size	A2 / Large
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P07-01, P08-01; Gate A0 active and verified; Alan review
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Dependencies	P07-01, P08-01
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	Business-model conversion trigger

1.4.2.1 Objective Create a deliberate managed-journey commitment with accepted terms, selected payment path, pinned config, and an enrollment invoice request independent from case state.

1.4.2.2 Scope

- Draft/accepted/cancelled/expired commitment transitions
- Payment-path choice
- Enrollment invoice command

1.4.2.3 Explicit non-goals

- No browser-triggered promotion
- No activation

1.4.2.4 Expected files or modules

- `app/domain/commitments.py`
- `app/storage/commitments.py`
- `app/routes/enrollment.py`
- `tests/python/test_commitments.py`

1.4.2.5 Change map

Area	Planned change
Schema/model	Commitment events/current projection.
API/routes	Create, accept, cancel, inspect enrollment.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.

Area	Planned change
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

1.4.2.6 Tests and validation Objective tests:

- Acceptance prerequisites
- duplicate accept
- stale config
- cancel/expire
- resume

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate **A2** on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

1.4.2.7 Diagnostics and observability Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

1.4.2.8 Rollout Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

1.4.2.9 Rollback or forward repair Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

1.4.2.10 Dependencies and parallel work Hard dependencies: P07-01, P08-01. Parallel opportunities: P07-04

1.4.2.11 Objective acceptance criteria

- ☐ Create a deliberate managed-journey commitment with accepted terms, selected payment path, pinned config, and an enrollment invoice request independent from case state.
- ☐ Every listed objective test passes locally and at the required gate where applicable.
- ☐ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- ☐ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.

☐ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

1.4.2.12 Ownership and review

Role	Recommendation
Work-package DRI	Luke
Primary implementer	Luke (proposed; pending Alan review)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
PR-level ownership exception	None
Required expertise	Python/FastAPI, PostgreSQL, security-aware product engineering
Higher-capability review	Mandatory before merge

1.4.3 P07-03 - Emit processor-neutral workspace promotion authorization

Field	Plan
Repository / base	`app` (`main`)
Gate / size	A2 / Large
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P04-02, P07-02, P08-02; Gate A0 active and verified; Alan review
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Dependencies	P04-02, P07-02, P08-02
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	Business-model promotion invariants

1.4.3.1 Objective Validate commitment, terms, commercial version, settled funding and credit evidence, then emit exactly one byos.workspace_promotion_authorized.v1 event; processor callbacks and browser redirects cannot emit it.

1.4.3.2 Scope

- Deterministic authorization predicate
- Canonical payload and idempotency key
- Transactional event/outbox
- Funding/credit/config evidence references

1.4.3.3 Explicit non-goals

- No case creation

- No redirect/webhook-as-authority
- No GC membership

1.4.3.4 Expected files or modules

- `app/domain/promotion_authorization.py`
- `app/event_handlers/commercial.py`
- `app/storage/outbox.py`
- `tests/python/test_promotion_authorization.py`

1.4.3.5 Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	Internal event producer only.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Only commercial-domain authority can emit the typed event.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

1.4.3.6 Tests and validation Objective tests:

- Pending/failed/reversed funding
- stale terms/config
- duplicate/replayed inputs
- credit mismatch
- event payload/checksum

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate **A2** on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

1.4.3.7 Diagnostics and observability Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

1.4.3.8 Rollout Deploy behind the narrowest practical feature/config gate, verify the named synthetic flow and diagnostics on a known revision, then widen only at the associated programme gate.

1.4.3.9 Rollback or forward repair Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

1.4.3.10 Dependencies and parallel work Hard dependencies: P04-02, P07-02, P08-02. Parallel opportunities: P07-01

1.4.3.11 Objective acceptance criteria

- ☐ Validate commitment, terms, commercial version, settled funding and credit evidence, then emit exactly one byos.workspace_promotion_authorized.v1 event; processor callbacks and browser redirects cannot emit it.
- ☐ Every listed objective test passes locally and at the required gate where applicable.
- ☐ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- ☐ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- ☐ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

1.4.3.12 Ownership and review

Role	Recommendation
Work-package DRI	Luke
Primary implementer	Luke (proposed; pending Alan review)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
PR-level ownership exception	None
Required expertise	event-driven systems, payments, commercial policy
Higher-capability review	Mandatory before merge

1.4.4 P07-04 - Promote workspace exactly once from authorization event

Field	Plan
Repository / base	`app` (`main`)
Gate / size	A2 / Exceptional
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P03-03, P04-02, P07-03; Gate A0 active and verified; Alan review
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Dependencies	P03-03, P04-02, P07-03

Field	Plan
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	byos.workspace_promotion_authorized.v1 contract, PR #124 source linkage concepts

1.4.4.1 Objective Consume byos.workspace_promotion_authorized.v1 to create/find one enrollment_pending paid case, link the source workspace, project allowlisted data, and grant only restricted IP membership.

1.4.4.2 Scope

- Unique promotion idempotency key
- Case/source link and config version
- Purpose-limited data manifest and audit event

1.4.4.3 Explicit non-goals

- No redirect authority
- No GC case membership transfer
- No activation

1.4.4.4 Expected files or modules

- `app/domain/promotion.py`
- `app/storage/promotions.py`
- `app/event_handlers/enrollment.py`
- `tests/python/test_workspace_promotion.py`

1.4.4.5 Change map

Area	Planned change
Schema/model	Promotion ledger and source-workspace unique constraint.
API/routes	Internal event handler only.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Projection defaults deny and preserves provenance.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

1.4.4.6 Tests and validation Objective tests:

- Replay/concurrency
- partial transaction

- funding event validation
- private candidate exclusion
- GC invitation required

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate **A2** on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

1.4.4.7 Diagnostics and observability Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

1.4.4.8 Rollout Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

1.4.4.9 Rollback or forward repair Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

1.4.4.10 Dependencies and parallel work Hard dependencies: P03-03, P04-02, P07-03. Parallel opportunities: P07-01, P07-02

1.4.4.11 Objective acceptance criteria

- ☐ Consume byos.workspace_promotion_authorized.v1 to create/find one enrollment_pending paid case, link the source workspace, project allowlisted data, and grant only restricted IP membership.
- ☐ Every listed objective test passes locally and at the required gate where applicable.
- ☐ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- ☐ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- ☐ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

1.4.4.12 Ownership and review

Role	Recommendation
Work-package DRI	Luke
Primary implementer	Luke (proposed; pending Alan review)

Role	Recommendation
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
PR-level ownership exception	None
Required expertise	event-driven systems, privacy, PostgreSQL
Higher-capability review	Mandatory before merge

1.5 Cross-repository coordination

Repository	Authority after merge
`app`	routing, validation, authorization, participant/case/workflow truth, PHI policy and commercial domain

Merge order inside this package: `P07-01` → `P07-02` → `P07-03` → `P07-04`.

Handoff rule: producer contract/version must merge and be pinned before its consumer enables the feature. During a partial merge, consumers fail closed or retain the last approved version; they do not fabricate data or silently broaden access.

Compatibility window: keep prior readers/writers and rollback image usable through the named gate; contract/migration PRs document the exact removal point.

1.6 Test and acceptance matrix

Test dimension	Method	Objective result
Happy path	Terms/config version pinning; Commitment retries; Exact-once promotion; Funding reversal handling; No GC access escalation	Expected state/event/UI and named evidence are produced once.
Authorization failure	Wrong/no subject, role, membership, purpose or consent	403/404 as appropriate; no write or data leak; audit safe failure.
Role differences	Separate IP, GC, operator, provider/service identities	Only the approved lens/projection/actions appear.
Retries/duplicates	Repeat same canonical request/event	Original result returned; no duplicate membership/case/invoice/document/event.
Concurrency	Two writers/claimants/migrators/workers	One valid transition; loser receives deterministic stale/conflict response.
Stale state	Old revision/config/terms/token/artifact	Fail closed or approved compatibility behavior; never silent coercion.

Test dimension	Method	Objective result
Malformed input	Boundary, encoding, size, enum, redirect and identifier corpus	Validation before side effects; safe error.
Unavailable dependency	Database, intelligence, processor, S3/scanner or auth unavailable	Explicit degraded/blocked behavior; no hallucinated success.
Partial failure	Crash between external effect, DB transaction and outbox	Retry/forward repair converges from durable evidence.
Logout/login/resume	Session expires/restarts mid-flow	Return to authorized current state; no query/link authority.
Mobile/accessibility	320/375/768 widths, keyboard, focus, reduced motion	Core task completes with truthful state and no clipped controls.
Synthetic cloud	Known dev revision and synthetic identities/data	Gate artifact records SHA, dependency status, audit and cleanup.
Audit/observability	Inspect events/metrics/logs	Versions, actor/authority, correlation and reason present; no sensitive payload.

1.7 Risks and decisions

Founder outcomes, Alan's remaining technical ownership and external activation dependencies are recorded separately so a final policy is never mistaken for an active control.

ID	Status	Classification	Remaining owner	Affected / latest	Programme effect
D-DATA-01	Founder decision final; implementation evidence pending	Luke decides product/privacy outcome	Alan - implementation	P04-02, P06-02, P06-03, P07-03, P07-04, P09-05, P10-01, P10-02; Before P04-02	Founder ambiguity is removed across WP04, WP06, WP07, WP09 and WP10; A1 still requires the technical denial evidence.
D-COM-01	Founder decision final for beta; tax externally dependent	Luke decides commercial outcome	Alan - implementation	P06-01, P07-01, P08-01; Before activation of a Complete configuration	P06-01, P07-01 and P08-01 may proceed synthetically; live activation remains gated.
D-COM-02	Founder decision final for v1; live activation gated	Luke decides commercial outcome	Alan - implementation	P07-01, P08-01, P08-03; Before P07-01 activation data	Synthetic commercial configuration and ledgers may proceed; real charging remains gated.

ID	Status	Classification	Remaining owner	Affected / latest	Programme effect
D-COM-03	Externally dependent; founder policy inputs settled	External specialist / later evidence required	External legal/privacy/finance; Luke final business approval	P07-01, P07-02, P08-03; Before P07-02 user acceptance	Synthetic A2 preparation may proceed; live P07-02/P08 charging and B1 customer acceptance remain blocked.
D-COM-04	Founder decision final; implementation evidence pending	Luke decides commercial/product outcome	Alan - implementation	P07-02, P07-03, P07-04, P08-03, P09-04; Before P07-02	P07-02 through P09-04 may proceed under the settled semantics.
D-COM-05	Founder decision final for beta	Luke decides beta scope	None during beta	P07-02, P08-02; Before financed option is exposed	Financing no longer blocks P08-02; the feature is deliberately out of scope.

1.7.1 D-DATA-01 - IP-GC field projection allowlist

Field	Recorded decision or requirement
Decision state	Final
Founder outcome	Use need-to-know sharing. Shared journey status and agreed relationship/outreach/logistics may cross roles; private IP notes, unconfirmed assertions and GC medical, reproductive, insurance, identity or wellbeing answers do not cross automatically. Provider packets require preview, consent and appropriate verification.
Founder constraints	Default-deny field-and-purpose allowlists; Source-labelled corrections; Named-purpose provider projection only
What remains for Alan	Define explicit field/purpose allowlists in code and projection contracts; Implement cross-role and provider negative tests
Evidence required	Cross-role denial matrix; Provider projection allowlist tests; Correction/source-label evidence
Approval rule	Luke approval is required for a broader sharing policy, not for compliant implementation detail.
External dependencies	None
Affected PRs / latest point	P04-02, P06-02, P06-03, P07-03, P07-04, P09-05, P10-01, P10-02; Before P04-02
Programme effect	Founder ambiguity is removed across WP04, WP06, WP07, WP09 and WP10; A1 still requires the technical denial evidence.

Field	Recorded decision or requirement
Independent recommendation	Default deny; expose only explicit purpose-limited projections. Class 1/2 and class 4 never automatically cross roles.

1.7.2 D-COM-01 - Node 0 Complete price

Field	Recorded decision or requirement
Decision state	Final beta policy; external tax wording pending
Founder outcome	Node 0 Complete is a voluntary $495 USD one-time upgrade paid by the IP or authorized IP account holder, never the GC, collected before Complete features and fully credited against the later Phase 1/JMF without gating matching or safety support.
Founder constraints	Transparent one-time feature/credit wording; Versioned prospective changes only; No forced payment to identify, approach or reach agreement with a GC
What remains for Alan	Implement configurable price, payer ownership, durable payment/credit ledger and idempotent flows
Evidence required	Synthetic success, failure, retry, duplicate and credit tests; Tax wording before live activation
Approval rule	Luke approves future commercial-policy changes; tax treatment requires specialist review.
External dependencies	Tax treatment and customer-facing wording before live activation
Affected PRs / latest point	P06-01, P07-01, P08-01; Before activation of a Complete configuration
Programme effect	P06-01, P07-01 and P08-01 may proceed synthetically; live activation remains gated.
Independent recommendation	Keep no active production value until explicitly approved; test with synthetic configuration.

1.7.3 D-COM-02 - Ratify JMF price book and effective date

Field	Recorded decision or requirement
Decision state	Final v1 policy; external activation dependencies
Founder outcome	Configuration v1 is $8,900 USD: a $750 deposit followed by $1,630, $2,445, $1,630, $1,630 and $815 milestones. Pay-in-full is $7,830 total, leaving $7,080 after the deposit; the separate $495 Complete credit must not be double-counted.
Founder constraints	Versioned configuration; Prospective changes only; First live beta requires A2 plus approved terms and tax treatment

Field	Recorded decision or requirement
What remains for Alan	Implement versioned configuration, milestone and reconciliation behavior
Evidence required	Synthetic schedule, pay-in-full and no-double-credit reconciliation tests
Approval rule	Luke approves future price-book versions and first live activation.
External dependencies	Approved terms and tax treatment before live activation
Affected PRs / latest point	P07-01, P08-01, P08-03; Before P07-01 activation data
Programme effect	Synthetic commercial configuration and ledgers may proceed; real charging remains gated.
Independent recommendation	Use the document anchors but require explicit version activation rather than code literals.

1.7.4 D-COM-03 - Terms, privacy and refund language versions

Field	Recorded decision or requirement
Decision state	Externally dependent
Founder outcome	No live acceptance or charge may rely on the contradictory current terms draft. D-COM-06 and D-COM-07 supply the approved business policy for a new versioned terms package.
Founder constraints	Resolve refundable-until-onboarding versus non-refundable contradiction; No retrospective policy changes; Versioned terms/privacy/refund acceptance
What remains for Alan	Implement versioned acceptance and keep real charging disabled until an approved version is active
Evidence required	Approved paid-service terms; Approved privacy/consent disclosure; Approved tax/refund/accounting wording
Approval rule	Luke approves the final business policy and customer-facing version after specialist review.
External dependencies	Legal, privacy, tax and finance review before real-user acceptance or charging
Affected PRs / latest point	P07-01, P07-02, P08-03; Before P07-02 user acceptance
Programme effect	Synthetic A2 preparation may proceed; live P07-02/P08 charging and B1 customer acceptance remain blocked.
Independent recommendation	External legal/privacy/finance review before any real acceptance.

1.7.5 D-COM-04 - Enrollment funding and activation evidence

Field	Recorded decision or requirement
Decision state	Final
Founder outcome	Use two-step commitment: settled $750 plus current accepted terms proves commercial enrollment, but payment alone never promotes. Exactly one enrollment-pending case follows explicit IP promotion and authenticated GC consent. The $1,630 activation tranche is earned only after defined IP/GC onboarding and screening coordination has begun; it never implies medical clearance.
Founder constraints	Redirects/webhooks never promote; Typed promotion event only; Source-labelled facts and explicit evidence checklist
What remains for Alan	Define typed events, activation checklist, idempotency and audit mechanics
Evidence required	Duplicate, replay, partial-failure, promotion and activation evidence tests
Approval rule	Luke approves any change to commercial enrollment or activation semantics.
External dependencies	None
Affected PRs / latest point	P07-02, P07-03, P07-04, P08-03, P09-04; Before P07-02
Programme effect	P07-02 through P09-04 may proceed under the settled semantics.
Independent recommendation	Funding confirmation is durable ledger evidence; promotion uses the sole typed authorization event; activation requires an explicit evidence checklist, actor and audit event.

1.7.6 D-COM-05 - Financing path

Field	Recorded decision or requirement
Decision state	Final beta scope
Founder outcome	Financing is excluded and hidden from beta; Natalyx does not advertise, recommend or integrate a lender or treat lender status as funding evidence.
Founder constraints	Card and ACH remain unaffected; Reconsider only after demand evidence plus legal/privacy review
What remains for Alan	Keep financing paths disabled and absent from beta UI/configuration
Evidence required	Negative configuration/UI tests proving financing is unavailable
Approval rule	A future financing proposal requires a new founder decision.
External dependencies	None
Affected PRs / latest point	P07-02, P08-02; Before financed option is exposed
Programme effect	Financing no longer blocks P08-02; the feature is deliberately out of scope.

Field	Recorded decision or requirement
Independent recommendation	Do not block card/ACH MVP; keep optional and disabled.

Implementation risk	Owner	Treatment
Configuration drift	Luke	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
Duplicate promotion	Luke	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
Redirect spoofing	Luke	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
Unresolved policy encoded as default	Luke	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.

1.8 Existing-work salvage

Source	Head	Valuable content	Destination	Recommendation	Extraction status/safety
No direct source	-	Build from approved contracts/evidence	Package PRs	New implementation	N/A

Package source inventory:

- Payment/business-model DOCX as authority
- PR #124 source workspace linkage concepts only

1.9 Ownership recommendation

Luke is the proposed DRI and primary implementer for the settled commercial contract. Alan reviews and retains the named technical-mechanism decisions; finance/legal review remains required where the live policy names it.

PR	WP DRI	Primary implementer	Technical-decision owner	Reviewer	Final approver	Exception
P07-01	Luke	Luke (proposed)	None; implement the settled founder constraints and approved contracts	Alan	Luke	None
P07-02	Luke	Luke (proposed)	None; implement the settled founder constraints and approved contracts	Alan	Luke	None
P07-03	Luke	Luke (proposed)	None; implement the settled founder constraints and approved contracts	Alan	Luke	None
P07-04	Luke	Luke (proposed)	None; implement the settled founder constraints and approved contracts	Alan	Luke	None

PR ownership inherits the proposed work-package split pending Alan's review. Named technical-decision ownership remains separate and never transfers implementation ownership across work packages; reciprocal and external review requirements remain intact.

1.10 Definition of done

- ☐ Every proposed PR is merged in dependency order or explicitly dispositioned, with its objective acceptance criteria and review record complete.
- ☐ All package blocking decisions are resolved by the named owner; no commercial/privacy/security value is guessed in code.
- ☐ Local, PostgreSQL, role/authorization, retry/concurrency, failure, mobile and synthetic-cloud evidence required by this package is attached.
- ☐ Cross-repository contracts and artifact/migration versions are pinned; partial-merge behavior is fail-closed or explicitly degraded.

- ☐ Observability is sufficient to identify revision, transition, dependency and safe failure without sensitive payloads.
- ☐ Existing PR/worktree value and reviewer findings are extracted before closure; no cleanup occurs without team confirmation.
- ☐ The associated programme gate is accepted by Luke and any required external reviewer; tests/deploy alone do not establish B1/B2.